LUKE P. CORCOS

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RESEARCH EXPERIENCE

Lawrence Berkeley National Laboratory — 2017 - 2023
Graduate Student Researcher

- Constructed high-fidelity mathematical and numerical models for multi-physics multi-layer coating flow dynamics, with applications to the leveling of multi-layer paint films.
- Developed a tailored dimension-independent high-performance computing (HPC) framework in C++, parallelized via MPI. Simulations were performed on NERSC's *Cori* and *Perlmutter* supercomputers; with up to 400,000 elements and 14 million spatial degrees of freedom.
- Collaborated with research scientists from the Lawrence Berkeley National Laboratory (LBNL) and PPG Industries Inc. via the Department of Energy's *HPC4Mfg* program.
- Performed a high-resolution parametric study at industrially relevant conditions motived from experiments on automobile paint coatings, providing actionable insights into the formation of irregularities and patterns within multi-layer paint coatings.

Other coding experience

- Wrote MPI-parallel block sparse linear algebra routines in C++, including: parallel matrix-matrix and matrix-vector multiply, quad/octree-based multigrid methods, preconditioned conjugate gradient and GMRES algorithms, Arnoldi eigenvalue solvers, and ILU decompositions.
- Implemented C++ algorithms for capturing rising bubble dynamics, including level set methods and projection methods for solving the incompressible Navier-Stokes equations.

MARKET EXPERIECE

Independent Market Researcher — 2016 - Present

- I actively trade stocks, options, and cryptocurrencies using the CANSLIM method along with self-discovered chart patterns. I operate a blog dedicated to stock trading: https://lpcorcos.github.io/stocks/.
- My compound annual growth rate (CAGR) since 2020 is 32%, with a Sharpe ratio of 0.72.
- In 2024, I returned 111%, outperforming the S&P 500 by 4.75x.

PUBLICATIONS

L.P. Corcos, R.I. Saye, J.A. Sethian, *A hybrid finite difference level set-implicit mesh discontinuous Galerkin method for multi-layer coating flows*. J. Comput. Phys. 2024, 507, 112960 (2024).

A.D. Cimmarusti, Z. Yan, B.D. Patterson, L.P. Corcos, L.A. Orozco, S. Deffner, *Environment-assisted speed-up of the field evolution in cavity quantum electrodynamics*, Physical Review Letters 114 (23), 233602 (2015).

EDUCATION

Ph.D. in Applied Mathematics — Received 2023
Awarded the Bernard Friedman Memorial Prize in Applied Mathematics.
Advisor: Professor James A. Sethian.
University of California, Berkeley.

B.S. in Mathematics and Physics — Received 2016
High Honors in Physics, Banneker Key Scholar, University Honors.
University of Maryland, College Park.

SKILLS

C++, MPI, MKL, LAPACK, Linux, Make, Bash, Mathematica, MatLab, LaTeX, Office.

References available upon request.